

Two-Dimensional Stabilization of Quartic del Pezzo Surfaces

Tavis Rudd*

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Abstract

Let k be a field of characteristic zero and S/k a smooth quartic del Pezzo surface. We prove that $S \times \mathbf{A}^2$ is k -rational if $S(k) \neq \emptyset$ and the Galois module $\mathrm{Pic}(S_{\bar{k}})$ is stably permutation. Thus every stably k -rational quartic del Pezzo surface becomes rational after adjoining two independent variables. The proof uses the projective Cox model of a universal torsor. A saturated rank-three subtorus has four Galois-stable groups of Cox generators whose weight differences form an integral basis. A codimension-three linear section containing a general tangent space meets a general subtorus orbit in one point. Generic tangent projection then identifies the rational quotient with projective four-space, and the remaining two-dimensional torus is rational. Applied to both families of Tschinkel and Zhang, the theorem gives $X \times \mathbf{P}^2$ rational for every cubic hypersurface in those families. For their cubic threefolds, a separate theorem on $X \times \mathbf{P}^1$ makes the stabilization index equal to two and yields smooth projective nonrational fourfolds Y such that $Y \times \mathbf{A}^1$ is rational.

Keywords. quartic del Pezzo surface; stable rationality; universal torsor; variety with one apparent double point; algebraic torus; rationality after affine-line stabilization.

1 Results

For a projective variety V/k , put

$$\ell_k(V) = \min\{m \geq 0 : V \times \mathbf{A}^m \text{ is } k\text{-rational}\}.$$

Since $\mathbf{A}^m$ and $\mathbf{P}^m$ are birational, one obtains the same integer with $\mathbf{P}^m$ in place of $\mathbf{A}^m$. Tschinkel and Zhang use this notion in their discussion of levels of stable rationality [3, p. 13].

Theorem 1.1 (Two-dimensional stabilization). *Let k be a field of characteristic zero and let S/k be a smooth quartic del Pezzo surface. Suppose that $S(k) \neq \emptyset$ and that $\mathrm{Pic}(S_{\bar{k}})$ is a stably permutation $\mathrm{Gal}(\bar{k}/k)$ -module. Then*

$$S \times \mathbf{A}^2 \quad \text{and, equivalently,} \quad S \times \mathbf{P}^2$$

are k -rational. In particular, this holds for every stably k -rational smooth quartic del Pezzo surface.

Tschinkel and Zhang prove stable rationality under these hypotheses by constructing a rational universal torsor and a stable-permutation resolution of its Néron–Severi torus [3, Corollary 4.3]. Their resolution gives the bound $S \times \mathbf{A}^{11}$ rational in the minimal cases.

*tavisrudd@damnsimple.com

Theorem 1.1 uses their Cox model and classification but replaces the eleven-dimensional quasi-trivial torus by a rational quotient and a two-dimensional torus. Their geometric results remain essential inputs.

For $r \geq 0$, let $X_{1,r}, X_{3,r} \subset \mathbf{P}_{\mathbf{Q}}^{r+4}$ be the smooth cubics

$$X_{1,r} : (x_4 - 2x_3)x_1^2 + 3(x_4 + 2x_3)x_2^2 + 3x_3^2x_4 - x_4^3 + x_5^3 + \sum_{i=1}^r w_i^3 = 0, \quad (1)$$

$$X_{3,r} : x_4(x_1^2 + 2x_1x_2) + x_3(x_1^2 + x_1x_2 + x_2^2) + x_3^3 - x_3^2x_4 + x_4^3 + 2x_5^3 + \sum_{i=1}^r w_i^3 = 0. \quad (2)$$

Corollary 1.2 (The Tschinkel–Zhang cubic families). *For every $r \geq 0$ and $j \in \{1, 3\}$, the variety $X_{j,r} \times \mathbf{P}^2$ is rational over $\mathbf{Q}$.*

Corollary 1.3 (Stabilization index and affine-line stabilization). *For $j \in \{1, 3\}$, let $X_j = X_{j,0} \subset \mathbf{P}_{\mathbf{Q}}^4$. Then*

$$\ell_{\mathbf{Q}}(X_j) = \ell_{\mathbf{C}}((X_j)_{\mathbf{C}}) = 2.$$

Moreover, $Y_j = X_j \times \mathbf{P}^1$ is a smooth projective nonrational fourfold over $\mathbf{Q}$, whereas $Y_j \times \mathbf{A}^1$ is rational over $\mathbf{Q}$.

Here the last assertion concerns birational rationality after taking a product with $\mathbf{A}^1$. It is distinct from the Zariski cancellation problem, which asks whether isomorphic affine cylinders have isomorphic bases.

2 A criterion for rational torus quotients

Let $Z \subset \mathbf{P}(V)$ be a geometrically integral n -fold over an infinite field k . At a smooth point $p \in Z(k)$, write

$$\pi_p : Z \dashrightarrow \mathbf{P}(V/\widehat{T}_p Z) \quad (3)$$

for projection from the embedded tangent space; the hat denotes the affine cone. In the application, $Z \subset \mathbf{P}^{2n+1}$ has one apparent double point, and π_p is birational for general p by [3, Theorem 2.4].

Theorem 2.1 (A quotient criterion). *Let a rank- r k -torus T act linearly and generically freely on $Z \subset \mathbf{P}(V)$, and suppose that the generic tangent projection π_p is birational to $\mathbf{P}^n$. Over k^s , suppose that V has nonzero weight spaces $V_{w_0}, \dots, V_{w_r}$ such that $w_1 - w_0, \dots, w_r - w_0$ is a $\mathbf{Z}$ -basis of $X^*(T_{k^s})$. Assume that a k -defined linear subspace $B \subset \mathbf{P}(V)$ contains all other weight spaces.*

Suppose that, for a general $p \in Z(k)$, there is a k -defined codimension- r linear subspace

$$\Lambda \supset \langle T_p Z, B \rangle$$

with equations $\lambda_1, \dots, \lambda_r$ such that, at a general $q \in Z$, the matrix

$$A(q)_{ij} = \lambda_i(q_j), \quad q_j \in V_{w_j}, \quad (4)$$

has rank r and all $r + 1$ maximal minors are nonzero. Then the Rosenlicht quotient Z/T is k -rational.

Proof. The image of Λ under π_p is a linear $\mathbf{P}^{n-r} \subset \mathbf{P}^n$, and

$$\pi_p|_{\Lambda \cap Z} : \Lambda \cap Z \dashrightarrow \mathbf{P}^{n-r} \quad (5)$$

is birational on the open where π_p is an isomorphism. It remains to show that $\Lambda \cap Z$ is a rational section of the T -action.

For a geometric point $q = \sum q_j + q_B$, where q_B is supported on the other weight spaces, define

$$\kappa_j(q) = (-1)^j \det A(q)_{\widehat{j}}, \quad 0 \leq j \leq r. \quad (6)$$

The equations of Λ vanish on B , so $tq \in \Lambda$ is equivalent to

$$A(q)(\chi^{w_0}(t), \dots, \chi^{w_r}(t))^t = 0.$$

Cramer's identity says that the kernel is generated by $(\kappa_0(q), \dots, \kappa_r(q))$. The basis hypothesis gives a unique $t(q) \in T(k^s)$, in projective coordinates, through

$$\chi^{w_j - w_0}(t(q)) = \frac{\kappa_j(q)}{\kappa_0(q)}, \quad 1 \leq j \leq r. \quad (7)$$

These are Laurent monomials in the displayed ratios. Uniqueness makes $t(q)$ Galois equivariant. Thus $s(q) = t(q)q$ is a k -rational, orbit-invariant retraction to $\Lambda \cap Z$ on the open where the minors do not vanish. Hence $\Lambda \cap Z \dashrightarrow Z/T$ is birational.

We record the inverse and its domain. Choose linear forms $\rho_0, \dots, \rho_{n-r}$ whose restrictions give homogeneous coordinates on the image of Λ in $\mathbf{P}(V/\widehat{T}_p Z)$. Let $U_p \subset Z$ be an open on which π_p is an isomorphism, and let $U \subset U_p$ be the further open where all κ_j , $\rho_0(s(q))$ is nonzero and the action is free. Then

$$\Phi : U/T \longrightarrow \mathbf{P}^{n-r}, \quad [q] \longmapsto [\rho_0(s(q)) : \dots : \rho_{n-r}(s(q))] \quad (8)$$

is the quotient parametrization. If $R_\alpha(x)$ are homogeneous equations for Z , the graph of its inverse over $\Phi(U/T)$ is the unique component dominating that open of

$$\begin{aligned} R_\alpha(x) &= 0 \quad \text{for every } \alpha, \\ \lambda_i(x) &= 0 \quad 1 \leq i \leq r, \\ y_j \rho_0(x) - y_0 \rho_j(x) &= 0 \quad 1 \leq j \leq n-r. \end{aligned} \quad (9)$$

Write its unique point over y as $x(y)$ and set $\Theta(y) = [x(y)]$. The last line gives $\Phi(\Theta(y)) = y$. Conversely, $s(q) \in \Lambda \cap U_p$ and has the same tangent-projection coordinates as $x(\Phi([q]))$; injectivity of $\pi_p|_{U_p}$ gives

$$\Theta(\Phi([q])) = [s(q)] = [q] \quad \text{in } U/T.$$

Thus both composites are the identity on the stated dense opens. In the Cox application, the R_α in (9) are the twenty quadrics in [3, Section 3]. Equations (7) and (9) give the rational map and its inverse without an unrecorded finite cover. $\square$

The integral-basis condition is essential. If the weight differences span a proper finite-index sublattice, the same equations generally define a finite cover of a T -orbit rather than a rational section.

3 The rank-three subtorus of the projective Cox model

Let T be the Néron–Severi torus, so $X^*(T_{\bar{k}}) = \text{Pic}(S_{\bar{k}})$. The Galois-invariant cocharacter

$$\lambda_{-K} : \mathbf{G}_m \longrightarrow T, \quad D \longmapsto (-K_S, D), \quad (10)$$

acts by simultaneous scalar multiplication on the sixteen Cox generators. Write $T_0 = T/\lambda_{-K}(\mathbf{G}_m)$, a torus of rank five. Over $\bar{k}$, the projective Cox model

$$Z = \text{Proj Cox}(S) \subset \mathbf{P}^{15}$$

is a seven-dimensional variety with one apparent double point [3, Lemma 3.2].

Use the marking of [3, Sections 3–4]. For the type I_3 Galois image, the sublattice

$$N_3 = \mathbf{Z}e_3 \oplus \mathbf{Z}e_4 \oplus \mathbf{Z}e_5 \subset X_*(T_{0,\bar{k}}) \quad (11)$$

is saturated and Galois stable. The two generators act on this basis by

$$\begin{pmatrix} -1 & 0 & 0 \\ -1 & 0 & 1 \\ -1 & 1 & 0 \end{pmatrix}, \quad \begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & -1 \\ 0 & 0 & -1 \end{pmatrix}.$$

Let $T_3 \subset T_0$ be the corresponding rank-three subtorus. Four Galois-stable groups of Cox generators have the following T_3 -weights:

weight	Cox generators
$(0, 1, 1)$	L_{13}, L_{23}, L_{35}
$(1, 0, 1)$	L_{14}, L_{24}, L_{45}
$(1, 1, 0)$	E_1, E_2, E_5
$(1, 1, 1)$	L_{12}, L_{15}, L_{25}

Every other Cox generator lies in the Galois-stable subspace

$$B = \mathbf{P}\langle E_3, E_4, L_{34}, Q \rangle. \quad (12)$$

The determinant of the three weight differences from $(0, 1, 1)$ is one.

Remark 3.1 (Saturation and the projective kernel). In the type- I_1 description, the three visible sign cocharacters in the last three coordinates are

$$(0, 1, 1), \quad (1, 0, 1), \quad (1, 1, 0).$$

They span the even-sum sublattice of $\mathbf{Z}^3$, of index two. Its saturation is $\mathbf{Z}^3$, obtained by adjoining

$$(1, 1, 1) = \frac{1}{2}((0, 1, 1) + (1, 0, 1) + (1, 1, 0)).$$

The map from the three parameter copies of $\mathbf{G}_m$ to their image in the projective torus has kernel μ_2 . The degree two obtained from the unsaturated parameters counts this kernel; it is not the degree of the Rosenlicht quotient. Saturation gives the integral-basis condition in Theorem 2.1.

Proposition 3.2 (A tangent section for the four Galois types). *For a smooth geometric quartic del Pezzo Cox model, there is a general tangent point p and a codimension-three linear subspace $\Lambda \supset \langle T_p Z, B \rangle$ which satisfies the maximal-minor condition of Theorem 2.1 for T_3 . If the Galois image has type I_0, I_1, I_2 , or I_3 , such a pair (p, Λ) exists over the ground field.*

Proof. Use the standard blowup points

$$(1 : 0 : 0), (0 : 1 : 0), (0 : 0 : 1), (1 : 1 : 1), (1 : a : b), \quad \Delta = ab(a-1)(b-1)(a-b) \neq 0. \quad (13)$$

For a tangent point with auxiliary coordinates $z = (z_1, z_2, z_3)$, the space of tangent hyperplanes containing B has dimension four. Evaluation on the four groups of Cox generators gives a 4-by-4 matrix. The following exact choices are used; $e = (e_1, \dots, e_5)$ and z' specify the second Cox point at which the evaluation matrix is tested.

i	z	e	z'
1	(1, 3, 7)	(2, 3, 5, 7, 11)	(2, 4, 9)
2	(2, 5, 11)	(3, 4, 7, 13, 17)	(1, 6, 10)
3	(3, 8, 13)	(5, 7, 11, 17, 19)	(2, 9, 15)
4	(4, 9, 17)	(2, 5, 11, 19, 23)	(3, 10, 18)

Let D_i be the determinant of this evaluation matrix, and let M_i be the fixed 8-by-8 Jacobian minor at the same tangent point. After discarding nonzero rational factors, their numerators are

$$\begin{aligned}
D_1 &= (8a - 27)^2(3968a - 1349b - 2619), \\
D_2 &= 568453977a^3 - 373279764a^2b - 1658788404a^2 + 59068867ab^2 \\
&\quad + 976763788ab + 925668016a - 128537024b^2 - 359163728b - 10185728, \\
D_3 &= 1643918400a^3 - 1105797420a^2b - 7261914240a^2 + 154718524ab^2 \\
&\quad + 3804690316ab + 9944961600a - 282623593b^2 - 3443769343b - 3454184244, \\
D_4 &= 35068545a^3 + 2864538a^2b - 236376549a^2 - 5350827ab^2 \\
&\quad + 24417420ab + 469658574a + 14536855b^2 - 90750394b - 214068162,
\end{aligned}$$

and

$$\begin{aligned}
M_1 &= -4(a-1)(b-7)(3a-b-2)(2ab+7a-9b), \\
M_2 &= -9(a-1)(2b-11)(3a-b-2)(4ab+11a-15b), \\
M_3 &= -25(a-1)(3b-13)(2a-b-1)(3ab+13a-16b), \\
M_4 &= -(a-1)(4b-17)(13a-5b-8)(32ab+85a-117b).
\end{aligned}$$

Thus witness i is valid precisely on the open $D_i M_i \neq 0$.

The cover is checked by eight cases. For $i = 1, 2, 3$, choose either $D_i = 0$ or $M_i = 0$. A lexicographic Gröbner calculation in $\mathbf{Q}[a, b, h]/(h\Delta - 1)$ gives

chosen equations	localized zero locus
$D_1 D_2 D_3, D_1 D_2 M_3, D_1 M_2 D_3, D_1 M_2 M_3$	empty
$M_1 D_2 D_3, M_1 D_2 M_3$	empty
$M_1 M_2 D_3$	$3a - b - 2 = Q_0(b) = 0$
$M_1 M_2 M_3$	$3a - b - 2 = (3b - 26)(3b - 13) = 0$

where $Q_0(b) = 31223016b^2 - 435944529b + 1306078948$. On the line $3a - b - 2 = 0$, the primitive numerator of $D_4 M_4$ is

$$-(b-1)^4(4b-17)(16b-85)Q_4(b), \quad Q_4(b) = 83246b^2 - 872181b + 2185995.$$

The evaluations

$$\begin{aligned}
Q_0(17/4) &= \frac{69121705}{4}, & Q_0(85/16) &= -\frac{4117757269}{32}, \\
Q_4(26/3) &= \frac{7918133}{9}, & Q_4(13/3) &= -\frac{272530}{9}.
\end{aligned}$$

exclude the linear factors. The quadratics are coprime by the Bézout identity

$$\begin{aligned}
&(2265746679974131615 - 377042650728395274b)Q_0(b) \\
&\quad + (141417109727495582904b - 1342668830289072756147)Q_4(b) \\
&= 24176690547344887359179755.
\end{aligned}$$

Consequently the four opens $D_i M_i \neq 0$ cover $\Delta \neq 0$. The displayed polynomials and all eight localized ideals are checked by exact arithmetic in the verification bundle.

It remains to descend a good pair. Fix a twist of one of the four types. Because T_3 and B are Galois stable, tangent points, three-frames of hyperplanes containing $\langle T_p Z, B \rangle$, and orbit test points form a geometrically integral incidence space over k . Rank, nonvanishing, smoothness, and birational tangent projection define a nonempty Galois-invariant open. Its projection to the tangent-point factor is dense constructible and therefore contains a nonempty open.

Choose the universal torsor whose fibre over a fixed point of $S(k)$ is trivial. Its k -points are Zariski dense by [3, Lemma 2.1], and the same holds in the projective Cox model. Hence the tangent-point open has a k -point. At that point, good three-frames form a nonempty open in an affine space over k , which has a k -point because k is infinite. No individual split-coordinate witness is asserted to descend.

For type I_3 , stability of N_3 , B , and the four groups of Cox generators is checked on the two displayed generators. Up to conjugacy in $W(D_5)$, the groups of types I_0, I_1, I_2 are subgroups of the group of type I_3 [3, Lemma 4.2]. Restriction and a change of marking preserve the same data. This exhausts the four Galois images allowed by the hereditary H^1 -condition; it is not a classification of arbitrary subgroups of $W(D_5)$. $\square$

4 Proofs of the rationality statements

Proof of Theorem 1.1. Suppose first that S is k -minimal. Since a stably permutation module has vanishing H^1 on every open subgroup, Proposition 4.1 and Lemma 4.2 of [3] place the Galois image in one of the four types used in Proposition 3.2. Lemma 3.2 and Theorem 2.4 of [3], together with Theorem 2.1, give

$$Z/T_3 \sim_k \mathbf{P}^4. \quad (14)$$

Let $\mathcal{T} \rightarrow S$ be a universal torsor under the Néron-Severi torus T , chosen to have a k -point above a fixed point of $S(k)$. The stable-permutation hypothesis and Hilbert's Theorem 90 give

$$H^1(F, T) = 0 \quad \text{for every field extension } F/k \quad (15)$$

by [3, Remark 2.2]. Thus the generic fibre over $k(S)$ has a point. Translation by the resulting rational section σ defines the T -equivariant birational map

$$S \times T \dashrightarrow \mathcal{T}, \quad (s, t) \mapsto \sigma(s)t. \quad (16)$$

The split scalar subtorus in (10) acts as common multiplication on every Cox coordinate. Quotienting (16) by it gives a T_0 -equivariant birational map $S \times T_0 \dashrightarrow Z$. Since $T_3 \subset T_0$ is a k -subtorus, Rosenlicht quotients give

$$Z/T_3 \sim_k S \times (T_0/T_3). \quad (17)$$

The quotient T_0/T_3 is a two-dimensional k -torus, hence is k -rational by [5]. Equations (14) and (17) prove that $S \times \mathbf{A}^2$ is k -rational.

If S is not k -minimal, a k -defined union of pairwise disjoint exceptional curves contracts to a del Pezzo surface S' of degree at least five. The image of a k -point of S is a k -point of S' , so S' , and therefore S , is k -rational by [4, Theorem 2.1].

Finally, stable rationality over the infinite field k supplies a k -point, and Manin's theorem makes $\text{Pic}(S_{\bar{k}})$ stably permutation [1]. This proves the last assertion. $\square$

Proof of Corollary 1.2. For $X = X_{j,r}$, Tschinkel and Zhang project from a plane containing a distinguished line. The base is $\mathbf{P}_{\mathbf{Q}}^{r+1}$, with function field $K = \mathbf{Q}(t_1, \dots, t_{r+1})$, and the generic cubic-surface fibre is K -birational, after contracting the distinguished line, to a smooth quartic del Pezzo surface S/K with a K -point. Their calculation identifies the Galois image as type I_1 for $j = 1$ and type I_3 for $j = 3$ [3, Propositions 5.1–5.2]. Theorem 1.1 gives

$$K(S)(u, v) \simeq K(z_1, z_2, z_3, z_4).$$

Since $\mathbf{Q}(X) = K(S)$ as extensions of K ,

$$\mathbf{Q}(X)(u, v) \simeq \mathbf{Q}(t_1, \dots, t_{r+1}, z_1, z_2, z_3, z_4)$$

is purely transcendental. This is rationality of $X \times \mathbf{A}^2$, equivalently of $X \times \mathbf{P}^2$. The argument is an identity of function fields; no specialization assertion is used. $\square$

Proof of Corollary 1.3. The one-stabilization theorem states that, for every smooth complex cubic threefold X , the variety $X \times \mathbf{P}^1$ is irrational [2, Theorem 1.1]. It applies to $(X_j)_{\mathbf{C}}$. Therefore neither X_j nor $X_j \times \mathbf{P}^1$ can be rational over $\mathbf{Q}$, because rationality survives extension of the ground field. Corollary 1.2 gives rationality at $m = 2$ over $\mathbf{Q}$, and hence also over $\mathbf{C}$. This proves both equalities of stabilization indices.

Set $Y_j = X_j \times \mathbf{P}^1$. It is nonrational, while

$$\mathbf{Q}(Y_j \times \mathbf{A}^1) = \mathbf{Q}(X_j)(u, t)$$

is purely transcendental by Corollary 1.2. Equivalently, $\mathbf{P}^1 \times \mathbf{A}^1$ is birational to $\mathbf{A}^2$. Hence $Y_j \times \mathbf{A}^1$ is rational. $\square$

5 Verification and scope

The cocharacter saturation, type- I_3 actions, Cox weights, determinant-one weight differences, and tangent-incidence cover use exact integer and rational arithmetic. The verification bundle checks all four pairs (D_i, M_i) , the eight localized branches, and the Bézout identity above with SymPy 1.14.0. A separate implementation checks the saturated lattice, the full type- I_3 action, the rank-two quotient lattice, and the displayed numerical tangent matrices. The quotient criterion, equivariant generic trivialization, incidence descent, and function-field argument are proved in the text and are not delegated to those programs.

No Lean development formalizes Theorem 1.1, Theorem 2.1, Proposition 3.2, or either corollary. The formal-coverage annotations therefore say **absent**.

The result uses Tschinkel–Zhang’s universal-torsor geometry and their classification of the four hereditary- H^1 Galois images. It does not reprove those results and does not classify all subgroups of $W(D_5)$. The bounded literature audit recorded with the manuscript found no displayed predecessor for the two-dimensional conclusion, but citation-graph closure was unavailable. Accordingly, the paper makes no global priority claim.

AI assistance disclosure

This project was developed with extensive assistance from OpenAI Codex and Anthropic Claude. Under the author’s direction, the systems assisted with proof exploration, literature research, exact computation, verification, and manuscript drafting and revision. The author checked the resulting arguments, computations, code, and cited sources, reviewed and edited all AI-assisted material, and assumes responsibility for all content.

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